Special Theory of Relativity

Concepts, Intuition, and Derivations

Merged notes: kinematics, dynamics, momentum–energy, four-momentum, Doppler effect, the Lorentz group and spacetime diagrams

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1 Basic Definitions

Frame: A coordinate system used to describe positions and times. **Observer:** The person/instrument measuring an event. **Event:** Something that happens at a specific point in space and time.

Inertial frame: at rest or moving with constant velocity (no acceleration). **Non-inertial frame:** an accelerating frame.

Relativity is about comparing measurements made by two inertial observers – one at rest (S) and one moving at constant velocity v (S') – of the same event.

2 Galilean Relativity (the "common sense" picture)

Before Einstein, everyone assumed space and time are absolute – a ruler has the same length and a clock ticks the same rate for every observer. This gives the **Galilean transformation** between frame S (at rest) and S' (moving with speed v along x):

$$x' = x - vt, \qquad y' = y, \qquad z' = z, \qquad t' = t \qquad (v \ll c)$$

Differentiating gives the velocity transformation:

$$u'_x = u_x - v, \qquad u'_y = u_y, \qquad u'_z = u_z$$

and differentiating again, acceleration (and hence force, since m is assumed constant) is the same in both frames:

$$a'_x = a_x, \quad a'_y = a_y, \quad a'_z = a_z \;\Rightarrow\; F'_x = F_x, \ldots$$

Intuition

If you walk forward at 2×10^8 m/s inside a rocket moving at 2×10^8 m/s, Galilean thinking says a stationary observer outside sees you moving at 4×10^8 m/s – simple vector addition. This works fine for cars and rockets, but it silently assumes velocities can add without limit.

2.1 Where Galilean relativity breaks down

Apply this logic to light. If frame S' moves at u relative to S, Galilean addition predicts the speed of light measured in S' should be $c' = c - u$. But every experiment (most famously Michelson–Morley) shows light always travels at c, in every inertial frame, regardless of the source's or observer's motion.

This is the crack that led Einstein, in 1905, to propose two postulates:

1) The laws of physics are the same in all inertial frames.

2) The speed of light c is the same in all inertial frames.

If (2) is true, then $x' = x - vt$, $t' = t$ cannot both be right – because that transformation is exactly what predicts $c' = c - v$. Something classical had to give: it turns out time and length are not absolute. They depend on the observer's speed.

3 Time Dilation

Setup (a "light clock"): A light pulse bounces vertically between two mirrors separated by distance L_0, inside a box.

Case 1 – box at rest

Light travels up and down a distance L_0 each way, so the round-trip time measured by an observer at rest with the box is

$$t_0 = \frac{2L_0}{c}.$$

Case 2 – box moving at speed v (as seen by an outside observer)

The outside observer sees the light pulse trace a zig-zag path (a triangle) because the box moves sideways by vt while the light is in transit, where t is the round-trip time they measure. Since light still moves at speed c for this observer, in half the round trip:

$$\left(\frac{ct}{2}\right)^2 = L_0^2 + \left(\frac{vt}{2}\right)^2$$

Intuition

The vertical distance the light must cover is still L_0, but now the light also has to cover horizontal ground because the box is moving. Since the light's speed c is fixed (postulate 2), covering extra horizontal distance in the same "up-down" motion means the whole trip must take longer. The speed limit on light is what forces time itself to stretch.

Solving for t:

$$t^2 = \frac{4L_0^2}{c^2}\left(1 - \frac{v^2}{c^2}\right)^{-1} \qquad \Longrightarrow \qquad t = \frac{t_0}{\sqrt{1 - v^2/c^2}}$$

Since $\sqrt{1 - v^2/c^2} < 1$ for any $v > 0$, we get $t \geq t_0$: the moving clock appears to run slow to the outside observer. This is **time dilation**. As a check: if $v = 0$, $t = t_0$ (no motion, no effect); as $v \to c$, $t \to \infty$.

$$t = \gamma t_0, \qquad \gamma \equiv \frac{1}{\sqrt{1 - v^2/c^2}}$$

3.1 Alternative derivation: directly from the Lorentz transformation

Time dilation can also be derived directly from the transformation of t itself, rather than from a light clock (this route is taken once the Lorentz transformation of Section 5 is available).

Setup: A clock is at rest in S', sitting at one fixed location x'. Two "ticks" of this clock are two events that happen at the same x', at times t'_1 and t'_2. Since both events occur at the same place in S', the time between them as measured by that co-located clock is the **proper time** $T_0 = t'_2 - t'_1$.

An outside observer in S watches the same two events but, because the clock is moving in S, the events happen at two different locations $x_1 \neq x_2$ in S. Using the inverse Lorentz transformation for time,

$$t = \gamma\left(t' + \frac{vx'}{c^2}\right),$$

and noting x' is the same for both events (the clock hasn't moved in its own frame):

$$t_1 = \gamma\left(t'_1 + \frac{vx'}{c^2}\right), \qquad t_2 = \gamma\left(t'_2 + \frac{vx'}{c^2}\right)$$

Subtracting, the common vx'/c^2 term cancels exactly, so writing $T = t_2 - t_1$,

$$T = \gamma(t_2' - t_1') = \gamma T_0 = \frac{T_0}{\sqrt{1 - v^2/c^2}}.$$

Intuition

The cancellation of the vx'/c^2 term is the key step, and it is only possible because both ticks happen at the same x'. This is precisely why "proper time" is special: it is the time interval measured by a clock that is present at both events, so there is no leftover simultaneity term to worry about. Any other observer, for whom the two events occur at different places, is forced to see a longer interval $T = \gamma T_0 \geq T_0$.

4 Length Contraction

Setup: A rod of rest length L_0 lies along the direction of motion, moving at speed v. A pulse of light is sent from one end to the other and back.

Case 1 – rod at rest

Round trip time: $t_0 = \dfrac{2L_0}{c}$.

Case 2 – rod moving, as seen by an outside observer

Let the length of the rod as measured by the outside observer be L (this is what we want to find). Since the rod moves in the same direction the light travels:

- **Forward trip:** light chases the far end, which recedes at v, so it covers relative distance at rate $c - v$: $t_1 = \dfrac{L}{c - v}$.
- **Return trip:** the near end approaches at v, so relative closing rate is $c + v$: $t_2 = \dfrac{L}{c + v}$.

Total time seen by the outside observer:

$$t = t_1 + t_2 = \frac{L}{c - v} + \frac{L}{c + v} = \frac{2Lc}{c^2 - v^2} = \frac{2L}{c\,(1 - v^2/c^2)}$$

Intuition

This "outside" time t is exactly the dilated time from Section 3, because it is the same light-clock experiment viewed by the same outside observer – just with the rod lying along the direction of motion instead of across it. So we can equate this t to $t = t_0/\sqrt{1 - v^2/c^2}$, and solve for L in terms of L_0.

Setting the two expressions for t equal:

$$\frac{2L_0}{c\sqrt{1 - v^2/c^2}} = \frac{2L}{c\,(1 - v^2/c^2)}$$

Solving for L:

$$L = L_0\sqrt{1 - \frac{v^2}{c^2}} = \frac{L_0}{\gamma}$$

Since $\sqrt{1 - v^2/c^2} < 1$, $L \leq L_0$: a moving rod appears shorter along its direction of motion. This is **length contraction**. (If $v = 0$, $L = L_0$; there is no contraction perpendicular to the motion.)

4.1 Alternative derivation: directly from the Lorentz transformation

Length contraction can also be derived by applying the Lorentz transformation itself to the two endpoints of the rod.

Setup: A rod is at rest in S' (say, inside a moving flight), with its ends at fixed positions x'_1 and x'_2. Because the rod does not move in S', the observer riding along with it can measure the two ends at any two (possibly different) times and still get the same, unambiguous proper length $L_0 = x'_2 - x'_1$.

The outside observer, in S, is watching a moving rod, so to get a sensible length they must record the positions of both ends at the *same instant* t of their own clock: $L = x_2 - x_1$ (measured at one common time t in S).

Intuition

This is the crux of the direct derivation: "length" only makes sense as "positions of both ends at the same time," and simultaneity itself is frame-dependent. The light-clock method sidesteps this by using a light pulse; the direct method confronts it head-on and is really a statement about relativity of simultaneity.

Apply the forward Lorentz transformation $x' = \gamma(x - vt)$ to each end, using the same t:

$$x'_1 = \gamma(x_1 - vt), \qquad x'_2 = \gamma(x_2 - vt)$$

Subtracting,

$$x'_2 - x'_1 = \gamma(x_2 - x_1) \quad \Longrightarrow \quad L_0 = \gamma L \quad \Longrightarrow \quad L = \frac{L_0}{\gamma}$$

which reproduces the light-clock result, now obtained purely from the coordinate transformation. The vt term is what makes this work: it is exactly the term that ties x'_1 and x'_2 to a single common t, and it is what would be missing (and give the wrong, expanding answer) if one instead held t' fixed and used the inverse transformation.

5 The Lorentz Transformation

We now need the actual coordinate transformation $x \to x'$ that replaces the Galilean one and is consistent with time dilation, length contraction, and a universal speed of light.

Ansatz: Since the transformation must be linear (uniform motion maps straight lines to straight lines) and reduce to the Galilean form for $v \ll c$, we guess a single unknown scale factor k (which cannot depend on position, but may depend on v):

$$x' = k(x - vt) \qquad \Longrightarrow \qquad x = k(x' + vt')$$

(the second equation follows by symmetry: from S''s point of view, S recedes at $-v$).

Finding k using the postulate that light travels at c in both frames: Send a light pulse from the common origin at $t = t' = 0$. In each frame its path obeys $x = ct$ and $x' = ct'$. Substituting into the two equations above and eliminating t, t', carrying through the algebra and solving for k gives

$$k = \gamma \equiv \frac{1}{\sqrt{1 - v^2/c^2}} \qquad \text{(the Lorentz/boost factor)}$$

Once $k = \gamma$ is known, substituting back and simplifying yields the transformation for t' as well. The complete Lorentz transformation is:

$$x' = \gamma(x - vt), \quad y' = y, \quad z' = z, \quad t' = \gamma\left(t - \frac{vx}{c^2}\right)$$

with inverse (swap primes and let $v \to -v$):

$$x = \gamma(x' + vt'), \qquad t = \gamma\left(t' + \frac{vx'}{c^2}\right)$$

Intuition

Compare to Galilean: $x' = x - vt$ is just stretched by $\gamma \geq 1$ – this is length contraction in disguise. And t' now depends on position x, not just on t: two events that are simultaneous in S (same t, different x) are not simultaneous in S'. This "relativity of simultaneity" is the deep new ingredient – it is what makes time dilation and length contraction mutually consistent rather than contradictory.

In matrix form, with $(x, y, z, t) \to (x', y', z', t')$:

$$\begin{pmatrix} x' \\ y' \\ z' \\ t' \end{pmatrix} = \begin{pmatrix} \gamma & 0 & 0 & -\gamma v \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -\gamma v/c^2 & 0 & 0 & \gamma \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ t \end{pmatrix}$$

6 Velocity Transformation (Einstein Addition)

Take differentials of the Lorentz transformation and divide:

$$u'_x = \frac{dx'}{dt'} = \frac{dx'/dt}{dt'/dt}, \qquad \frac{dx'}{dt} = u_x - v, \qquad \frac{dt'}{dt} = \gamma\left(1 - \frac{v}{c^2}u_x\right)$$

$$u'_x = \frac{u_x - v}{1 - \dfrac{vu_x}{c^2}}, \qquad u'_y = \frac{u_y\sqrt{1 - v^2/c^2}}{1 - \dfrac{vu_x}{c^2}}, \qquad u'_z = \frac{u_z\sqrt{1 - v^2/c^2}}{1 - \dfrac{vu_x}{c^2}}$$

Inverse (replace $v \to -v$, swap primes):

$$u_x = \frac{u'_x + v}{1 + \dfrac{vu'_x}{c^2}}$$

Intuition

Compare to naive Galilean addition $u'_x = u_x - v$: the new formula has an extra factor $1 - vu_x/c^2$ in the denominator that throttles the result so it can never exceed c. Check: if $u_x = c$ (a light ray), then $u'_x = \dfrac{c - v}{1 - v/c} = c$ – light is measured at speed c by every observer, exactly as required. This is Einstein's velocity addition rule, and for everyday speeds ($v, u_x \ll c$) the denominator ≈ 1 and it reduces back to ordinary Galilean addition.

For three frames A, B, C chained together, the same idea generalizes to

$$v_{AC} = \frac{v_{AB} + v_{BC}}{1 + \dfrac{v_{AB} v_{BC}}{c^2}}.$$

7 The Invariant Interval and Minkowski Space

Define the spacetime interval between two events:

$$ds^2 = c^2 dt^2 - dx^2 - dy^2 - dz^2$$

A direct (if tedious) substitution of the Lorentz transformation shows

$$ds^2 = c^2 dt^2 - dx^2 - dy^2 - dz^2 = c^2 dt'^2 - dx'^2 - dy'^2 - dz'^2 = ds'^2$$

Intuition

Ordinary rotations in a plane leave the length of a vector, $x^2 + y^2$, unchanged even though the individual components x, y change – that's just $R^T R = I$. The Lorentz transformation is analogous: it "rotates" space and time into each other, but leaves the combination ds^2 unchanged. ds^2 plays the same role in spacetime that ordinary length plays in space – it's the one thing all inertial observers agree on, even though they disagree on t, x, y, z separately.

This motivates defining a 4-dimensional Minkowski space with coordinates $x_0 = ict$, $x_1 = x$, $x_2 = y$, $x_3 = z$ so that $ds^2 = -\left(x_0^2 + x_1^2 + x_2^2 + x_3^2\right)$ looks like an ordinary (Euclidean) distance – or, in the more common modern convention, a Lorentz vector A^μ ($\mu = 0, 1, 2, 3$, with 0 the temporal and $1, 2, 3$ the spatial components) has an associated Lorentz scalar

$$A_\mu A^\mu = A_0 A^0 + A_1 A^1 + A_2 A^2 + A_3 A^3 = A_0^2 - A_1^2 - A_2^2 - A_3^2$$

(using the raising/lowering prescription $A_i \to A^i = -A_i$, $A_0 \to A^0 = A_0$) which is invariant under Lorentz transformations, just as $x^2 + y^2 + z^2$ is invariant under ordinary rotations. Any two Lorentz vectors A^μ, B^μ similarly give invariant scalars $A_\mu A^\mu$, $B_\mu B^\mu$, $A_\mu B^\mu$.

8 Generalized Lorentz Transformation (Arbitrary Direction of $\vec{v}$)

So far $\vec{v}$ was assumed to lie along the x-axis. If $\vec{v}$ points in an arbitrary direction, split the position vector into a part parallel to $\vec{v}$ and a part perpendicular to it:

$$\vec{r} = \vec{r}_\parallel + \vec{r}_\perp, \qquad \vec{r}_\parallel = \frac{(\vec{v} \cdot \vec{r})}{v^2} \vec{v}$$

Intuition

Length contraction only occurs along the direction of relative motion; nothing contracts perpendicular to it. So treat $\vec{r}_\parallel$ exactly like the 1D case (contract it, mix it with t) and leave $\vec{r}_\perp$ untouched – this is precisely why $y' = y$, $z' = z$ when $\vec{v}$ happened to point along x.

Applying the known 1D result to the parallel component only:

$$\vec{r}'_\parallel = \gamma(\vec{r}_\parallel - \vec{v}t), \qquad \vec{r}'_\perp = \vec{r}_\perp, \qquad t' = \gamma\left(t - \frac{\vec{v}\cdot\vec{r}}{c^2}\right)$$

Recombining $\vec{r}' = \vec{r}'_\perp + \vec{r}'_\parallel = \vec{r} + (\gamma-1)\vec{r}_\parallel - \gamma\vec{v}t$ and substituting $\vec{r}_\parallel$:

$$\vec{r}' = \vec{r} + (\gamma-1)\frac{(\vec{v}\cdot\vec{r})}{v^2}\vec{v} - \gamma\vec{v}t, \qquad t' = \gamma\left(t - \frac{\vec{v}\cdot\vec{r}}{c^2}\right)$$

8.1 The full 4×4 matrix

Writing $\vec{v} = (v_x, v_y, v_z)$ and $v^2 = v_x^2 + v_y^2 + v_z^2$, the transformation $(x, y, z, t) \to (x', y', z', t')$ is $X' = \Lambda X$ with

$$\Lambda_{ij} = \delta_{ij} + (\gamma-1)\frac{v_i v_j}{v^2}\ (i,j = x,y,z), \qquad \Lambda_{i0} = -\gamma v_i, \qquad \Lambda_{0j} = -\frac{\gamma v_j}{c^2}, \qquad \Lambda_{00} = \gamma$$

Explicitly:

$$\begin{pmatrix} x' \\ y' \\ z' \\ t' \end{pmatrix} = \begin{pmatrix} 1+(\gamma-1)\frac{v_x^2}{v^2} & (\gamma-1)\frac{v_x v_y}{v^2} & (\gamma-1)\frac{v_x v_z}{v^2} & -\gamma v_x \\ (\gamma-1)\frac{v_y v_x}{v^2} & 1+(\gamma-1)\frac{v_y^2}{v^2} & (\gamma-1)\frac{v_y v_z}{v^2} & -\gamma v_y \\ (\gamma-1)\frac{v_z v_x}{v^2} & (\gamma-1)\frac{v_z v_y}{v^2} & 1+(\gamma-1)\frac{v_z^2}{v^2} & -\gamma v_z \\ -\frac{\gamma v_x}{c^2} & -\frac{\gamma v_y}{c^2} & -\frac{\gamma v_z}{c^2} & \gamma \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ t \end{pmatrix}$$

where $\gamma = 1/\sqrt{1 - v^2/c^2}$ as before.

8.2 Check: reduces to the earlier matrix when $\vec{v} = (v, 0, 0)$

Set $v_y = v_z = 0$, $v_x = v$. Then $v_i v_j / v^2$ is nonzero only for $i = j = x$ (equal to 1), so every cross term $(xy, xz, yz, \ldots)$ vanishes:

$$\Lambda_{xx} = 1 + (\gamma-1)(1) = \gamma, \qquad \Lambda_{yy} = \Lambda_{zz} = 1, \qquad \Lambda_{x0} = -\gamma v, \qquad \Lambda_{0x} = -\frac{\gamma v}{c^2}$$

which gives back exactly the x-only boost matrix of Section 5. So the x-only boost is not a separate case; it is simply this general matrix with $\vec{v}$ pointing along one axis, which is why every off-diagonal spatial entry collapses to zero except the single xx term.

9 Relativistic Mass

The Lorentz transformation fixes how space and time coordinates change between inertial frames. The next question is what happens to dynamics – momentum, mass, force, and energy – once

velocities no longer add the ordinary (Galilean) way. If momentum is still defined as $\vec{p} = m\vec{v}$ with m a fixed constant, momentum conservation in one frame does not guarantee momentum conservation in another frame related by a Lorentz boost. Something in the definition of momentum has to change. It turns out that the fix is to let the "mass" of a moving object depend on its speed.

Intuition

This mirrors exactly how time dilation and length contraction worked: Einstein's postulates broke the Galilean picture of absolute time and length, forcing t and L to become observer-dependent. Here, demanding that momentum conservation survive a Lorentz boost breaks the Galilean picture of a speed-independent mass, forcing m to become observer- (i.e. speed-) dependent instead.

9.1 Derivation 1: from a symmetric collision and velocity addition

Setup: Consider two identical particles of rest mass m_0 colliding and sticking together (a perfectly inelastic collision), analyzed in two frames related by the velocity-addition rules already derived. Let particle 1 have velocity u' and particle 2 have velocity $-u'$ in a frame S' that itself moves at velocity v relative to the lab frame S. Requiring momentum to be conserved in both frames,

$$m_1\vec{v}_1 + m_2\vec{v}_2 = (m_1 + m_2)\vec{v},$$

where $\vec{v}_1, \vec{v}_2$ are the particle velocities as seen in S, obtained from u' and v by the Einstein velocity-addition rule

$$v_1 = \frac{u' + v}{1 + \dfrac{u'v}{c^2}}, \qquad v_2 = \frac{-u' + v}{1 - \dfrac{u'v}{c^2}}.$$

Intuition

The trick is the same one used to find the Lorentz factor γ in Section 5: pick the simplest possible physical process (here, a symmetric collision) and demand that the same conservation law hold after transforming velocities with the (already known) relativistic velocity-addition formula. Whatever extra factor is needed to make that work is the definition of relativistic mass.

Write $x \equiv u'v/c^2$ for brevity. Substituting into the conservation equation and collecting terms in u' and v separately gives

$$\frac{m_1}{m_2} = \frac{1 + x}{1 - x}, \qquad x = \frac{u'v}{c^2}$$

Using v_1 and v_2 above, one can show directly that

$$c^2 - v_1^2 = \left(c^2 - u'^2\right)\left(1 - \frac{v^2}{c^2}\right)(1 + x)^{-2}, \qquad c^2 - v_2^2 = \left(c^2 - u'^2\right)\left(1 - \frac{v^2}{c^2}\right)(1 - x)^{-2}.$$

Dividing these two expressions cancels the common factor $(c^2 - u'^2)(1 - v^2/c^2)$ and leaves exactly the combination $\dfrac{1 + x}{1 - x}$ found above:

$$\sqrt{\frac{c^2 - v_2^2}{c^2 - v_1^2}} = \frac{1 + x}{1 - x} = \frac{m_1}{m_2} \qquad \Longrightarrow \qquad \frac{m_1}{m_2} = \frac{\sqrt{1 - v_2^2/c^2}}{\sqrt{1 - v_1^2/c^2}}.$$

Rearranging so that each side depends on only one particle,

$$m_1\sqrt{1-\frac{v_1^2}{c^2}} = m_2\sqrt{1-\frac{v_2^2}{c^2}} = m_0 = \text{constant}.$$

Intuition

Both sides of this equation refer to different particles with different speeds, yet they are equal. The only way that is possible is if the quantity $m\sqrt{1-v^2/c^2}$ is the same number for every particle in the problem – a universal constant, independent of speed. That constant, m_0, is what we call the **rest mass**: the mass measured in a frame where the particle is at rest ($v = 0$).

This immediately gives the mass of a particle moving at speed v, as measured by an observer relative to whom it moves:

$$m = \frac{m_0}{\sqrt{1-v^2/c^2}} = \gamma m_0$$

Here m is the relativistic mass, m_0 is the rest mass, and v is the speed of the particle relative to the observer. Since $\gamma \geq 1$ always, $m \geq m_0$, with equality only when $v = 0$. This is the dynamical counterpart of $t \geq t_0$ (time dilation) and complements $L \leq L_0$ (length contraction): nothing about the object itself has changed, but its inertia, its clock rate, and its length as measured by a relatively moving observer all shift together, governed by the same factor γ.

9.2 Derivation 2: Tolman's symmetric-collision argument

A second thought experiment (essentially Tolman's argument) pins down the same mass–velocity relation from a different angle.

Setup: Consider a perfectly inelastic collision – two identical lumps of matter collide and stick together. **As seen by an observer riding along with the collision** (the center-of-momentum frame): the two lumps, each of rest mass m_0, approach symmetrically with equal and opposite velocities $+v'$ and $-v'$, collide, and come to rest as one combined lump (all kinetic energy is absorbed; total momentum in this frame is zero both before and after, consistent with conservation).

Intuition

Choosing the perfectly symmetric case is what makes the thought experiment solvable: by symmetry neither lump can be treated as "special," so whatever inertia (mass) each carries must be the same function of speed for both. This lets us isolate how mass depends on speed without already knowing the answer.

As seen by an outside observer S, relative to whom this whole center-of-momentum frame itself drifts at some small speed v (along the direction of v'): the two lumps now have different speeds, because one moves with v' and the frame drift v, while the other moves with $-v'$ and the frame drift v. Using Einstein's velocity addition formula,

$$v_1 = \frac{v'+v}{1+\dfrac{v'v}{c^2}}, \qquad v_2 = \frac{-v'+v}{1-\dfrac{v'v}{c^2}}$$

so in S the lump moving with v_1 is faster than the one moving with v_2. After the collision, the combined lump moves at the common velocity $V = v$ (the velocity of the center-of-momentum frame itself, as measured in S).

Let m_1 and m_2 be the (velocity-dependent) masses of the two lumps as measured in S, and let M be the mass of the combined lump. Conservation of momentum in S requires

$$m_1 v_1 + m_2 v_2 = MV = (m_1 + m_2)V$$

(mass is also conserved in a perfectly inelastic collision, so $M = m_1 + m_2$). Taking the limit $v \to 0$ – i.e. letting the collision frame drift only infinitesimally relative to S – and working the momentum-balance equation out to first order in v reduces it to a direct comparison between m_1 and m_2:

$$\frac{m_1}{m_2} = \frac{dt_2}{dt_1}$$

i.e. the ratio of masses equals the ratio of the time-dilation factors for the two lumps' motion. Carrying this through with the Lorentz factor for each lump's speed gives the same standard result: a lump with rest mass m_0, moving at speed u relative to an observer, has measured (inertial) mass

$$m(u) = \frac{m_0}{\sqrt{1 - u^2/c^2}} = \gamma_u m_0$$

Intuition

This is the same γ factor as time dilation and length contraction, now attached to inertia itself: the faster an object moves relative to you, the more sluggish (harder to accelerate further) it appears, diverging as $u \to c$. This is precisely why no massive object can be accelerated up to c – its effective inertia grows without bound as it approaches the speed of light, so no finite force can finish the job. It also reconciles momentum conservation with Einstein's velocity addition: relativistic momentum $p = m(u)u = \gamma_u m_0 u$ is the quantity that is actually conserved in collisions, not the naive Newtonian $m_0 u$.

10 Relativistic Momentum

With $m = \gamma m_0$, the sensible generalization of momentum that is conserved in every inertial frame is

$$\vec{p} = m\vec{v} = \gamma m_0 \vec{v} = \frac{m_0 \vec{v}}{\sqrt{1 - v^2/c^2}}$$

rather than the Newtonian $\vec{p} = m_0 \vec{v}$. As $v \to c$, $p \to \infty$: an infinite force would be required to push any massive object up to the speed of light, which is why c acts as a hard speed limit.

11 Relativistic Kinetic Energy

11.1 Definition

Kinetic energy is still defined as the work done to accelerate a particle from rest to speed v:

$$T = \int_0^v F\,dx = \int \frac{dp}{dt}\, v\,dt = \int_0^v v\,d(mv) = \int_0^v v\,d(\gamma m_0 v) = m_0 \int_0^v v\,d(\gamma v).$$

Intuition

This is exactly the Newtonian definition of work, $T = \int F\,dx$, with $F = dp/dt$. Nothing about the definition of kinetic energy changes – what changes is that $p = \gamma m_0 v$ now carries the extra γ factor, so the integral works out differently than the familiar $\frac{1}{2}m_0v^2$.

11.2 Evaluating the integral

Since $\gamma v = v\left(1 - v^2/c^2\right)^{-1/2}$, differentiate using the product rule:

$$\frac{d(\gamma v)}{dv} = \left(1 - \frac{v^2}{c^2}\right)^{-1/2} + \frac{v^2}{c^2}\left(1 - \frac{v^2}{c^2}\right)^{-3/2} = \left(1 - \frac{v^2}{c^2}\right)^{-3/2} = \gamma^3.$$

So $d(\gamma v) = \gamma^3\,dv$, and the kinetic energy becomes

$$T = m_0 \int_0^v \gamma^3 v\,dv = m_0 \int_0^v \frac{v\,dv}{(1 - v^2/c^2)^{3/2}}.$$

Substitute $u = 1 - v^2/c^2$, so $du = -\frac{2v}{c^2}dv \Rightarrow v\,dv = -\frac{c^2}{2}du$:

$$T = -\frac{m_0c^2}{2}\int u^{-3/2}\,du = -\frac{m_0c^2}{2}\left(-2u^{-1/2}\right) = m_0c^2\,u^{-1/2}\Big|_0^v.$$

Evaluating the limits ($u^{-1/2} = 1$ at $v = 0$):

$$T = m_0c^2\left(\frac{1}{\sqrt{1 - v^2/c^2}} - 1\right).$$

$$T = m_0c^2(\gamma - 1)$$

Intuition

Check the low-speed limit: for $v \ll c$, a Taylor expansion gives $\gamma \approx 1 + \frac{1}{2}v^2/c^2$, so $T \approx m_0c^2 \cdot \frac{1}{2}v^2/c^2 = \frac{1}{2}m_0v^2$ – exactly the familiar Newtonian kinetic energy. The relativistic formula does not replace the old one; it contains it as the $v \ll c$ approximation, with corrections that only matter as $v \to c$.

12 Mass–Energy Equivalence

Since $m = \gamma m_0$, the kinetic energy result $T = m_0c^2(\gamma - 1)$ can be rewritten as

$$T = \gamma m_0c^2 - m_0c^2 = mc^2 - m_0c^2.$$

Rearranging, $mc^2 = T + m_0c^2$. Define the total relativistic energy $E \equiv mc^2$. Then

$$E = T + m_0c^2$$

Quantity	Meaning
$E = mc^2$	total relativistic energy
T	kinetic energy
m_0c^2	rest energy (energy present even when $v = 0$)

Intuition

Even a particle sitting perfectly still ($T = 0$) carries energy $E = m_0c^2$ simply by virtue of having mass. Mass is not a separate quantity from energy – it is a form of energy. This is the origin of the famous statement that mass and energy are interchangeable: converting a small amount of rest mass into kinetic energy or radiation (as in nuclear reactions) releases an amount of energy $\Delta E = \Delta m_0\, c^2$, and the huge size of c^2 is why tiny mass changes correspond to enormous energy release.

13 The Energy–Momentum Relation

Since $E = \gamma m_0 c^2$ and $p = \gamma m_0 v$, eliminate v between them. Multiply p by c:

$$pc = \gamma m_0 vc \quad \Longrightarrow \quad E^2 - (pc)^2 = \gamma^2 m_0^2 c^4 - \gamma^2 m_0^2 v^2 c^2 = \gamma^2 m_0^2 c^2 \left(c^2 - v^2\right).$$

Factor out c^2 inside the bracket:

$$E^2 - (pc)^2 = \gamma^2 m_0^2 c^4 \left(1 - \frac{v^2}{c^2}\right) = m_0^2 c^4 \cdot \underbrace{\gamma^2 \left(1 - \frac{v^2}{c^2}\right)}_{=1} = m_0^2 c^4,$$

using $\gamma^2 = \left(1 - v^2/c^2\right)^{-1}$, so the bracket exactly cancels the γ^2. Hence

$$E^2 = p^2c^2 + m_0^2c^4$$

Intuition

This is the dynamical analogue of the invariant interval $ds^2 = c^2dt^2 - dx^2 - dy^2 - dz^2$ from Section 7: just as ds^2 is the combination of t and $\vec{r}$ that every inertial observer agrees on, $m_0^2c^4 = E^2 - (pc)^2$ is the combination of energy and momentum that every observer agrees on, even though E and $\vec{p}$ separately look different in different frames. It also shows immediately why light ($m_0 = 0$) must always travel at exactly c: setting $m_0 = 0$ gives $E = pc$, and combined with $E = \gamma m_0 c^2$ this is only finite/nonzero in the limit $v \to c$.

14 Summary

- Galilean relativity assumes absolute space and time; it fails for light.
- Einstein's postulates (same physics, same c, in every inertial frame) force time, length, and mass to become observer-dependent.
- **Time dilation:** $t = \gamma t_0$ – moving clocks run slow.
- **Length contraction:** $L = L_0/\gamma$ – moving rods (along motion) shrink.
- The **Lorentz transformation** replaces the Galilean one and mixes space and time together; its general form for arbitrary $\vec{v}$ reduces to the x-axis boost as a special case.
- **Velocity addition** is capped so nothing ever exceeds c.

- The interval $ds^2 = c^2dt^2 - dx^2 - dy^2 - dz^2$ is the invariant that survives, playing the role that "distance" played in pre-relativistic physics.
- **Relativistic mass:** $m = \gamma m_0$, with $m \geq m_0$ always – derivable both from symmetric-collision velocity addition and from Tolman's argument.
- **Relativistic momentum:** $\vec{p} = \gamma m_0 \vec{v}$, which diverges as $v \to c$.
- **Relativistic kinetic energy:** $T = m_0c^2(\gamma - 1)$, reducing to $\frac{1}{2}m_0v^2$ for $v \ll c$.
- **Mass–energy equivalence:** $E = mc^2 = T + m_0c^2$; rest mass itself is a store of energy.
- **Energy–momentum invariant:** $E^2 = p^2c^2 + m_0^2c^4$, the dynamical counterpart of the spacetime interval ds^2.

Part II

Four-Momentum, Doppler Effect, the Lorentz Group, and Spacetime Diagrams

Part I ended with $E^2 = p^2c^2 + m_0^2c^4$. Everything below picks up from there and follows the order of the class notes (28/09, 29/09, 05/10, 06/10). Sign convention used throughout unless stated: metric $g_{\mu\nu} = \mathrm{diag}(+1, -1, -1, -1)$, $x^\mu = (ct, x, y, z)$, $\beta = v/c$, $\gamma = 1/\sqrt{1-\beta^2}$.

15 Four-Momentum

The energy–momentum relation says that E and $\vec{p}$ are not independent: they combine into a single object with the same transformation behaviour as $(ct, \vec{x})$.

$$p^\mu = (p^0, \vec{p}), \qquad p^0 = \frac{E}{c}.$$

Invariant "length". Lower the index with $g_{\mu\nu}$:

$$\begin{aligned} p^\mu p_\mu &= p^0 p_0 + p^i p_i = (p^0)^2 - \vec{p}^{\,2} \\ &= \frac{E^2}{c^2} - \vec{p}^{\,2} \\ &= \frac{(p^2c^2 + m_0^2c^4) - p^2c^2}{c^2} \quad (\text{using } E^2 = p^2c^2 + m_0^2c^4) \\ &= m_0^2c^2. \end{aligned}$$

$$p^\mu p_\mu = m_0^2c^2 \qquad \text{(a Lorentz scalar, same in every inertial frame)}$$

Intuition

$x^\mu x_\mu = c^2t^2 - \vec{x}^{\,2}$ is the invariant "length" of a position four-vector; $p^\mu p_\mu$ is the invariant "length" of the momentum four-vector, and its value is simply the rest mass (times c). E and $\vec{p}$ individually change from frame to frame, exactly like t and $\vec{x}$, but the combination $E^2/c^2 - p^2$ does not. This is Section 13 re-written in the language of four-vectors.

Useful consequences.

- *Relation to four-velocity.* With $u^\mu = \gamma(c, \vec{v})$ we get $p^\mu = m_0 u^\mu = (\gamma m_0 c, \gamma m_0 \vec{v}) = (E/c, \vec{p})$, consistent with $E = \gamma m_0 c^2$ and $\vec{p} = \gamma m_0 \vec{v}$.
- *Transformation under a boost along x* (same matrix as for x^μ):

$$\frac{E'}{c} = \gamma\Big(\frac{E}{c} - \beta p_x\Big), \qquad p'_x = \gamma\Big(p_x - \beta\frac{E}{c}\Big), \qquad p'_y = p_y,\ p'_z = p_z.$$

- *Photon* ($m_0 = 0$): $p^\mu p_\mu = 0$, so $E = pc$ and p^μ is a null vector. With $E = h\nu$ this is the bridge to the Doppler effect below.

16 Doppler Effect for Light

Sound needs a medium, so its Doppler shift depends on who moves relative to the air. Light has no medium and c is the same for all observers, so the shift must depend only on the *relative* velocity, and it must include time dilation.

Notation. $\nu_0 = 1/t_0$ is the frequency in the source's rest frame (t_0 = proper period). $\nu = 1/T$ is the frequency the observer actually receives.

16.1 Transverse case (observer perpendicular to the motion)

When the relative motion is perpendicular to the line of sight, there is no change in the light travel distance between successive wave crests. The only effect is time dilation: the observer sees the source's period stretched to $t = \gamma t_0$, so

$$\nu = \frac{1}{t} = \frac{\nu_0}{\gamma} = \nu_0\sqrt{1 - \frac{v^2}{c^2}} \;<\; \nu_0.$$

(This is the page in the notes with the observer drawn as a vertical line and waves arriving horizontally.)

16.2 Case 1: source receding from the observer (red shift)

The source moves *away* from the observer, so between two successive crests the source has moved a further distance vt away, and the light needs an extra time vt/c to cover it.

$$T = t + \frac{vt}{c} = t\Big(1 + \frac{v}{c}\Big), \qquad t = \gamma t_0 = \frac{t_0}{\sqrt{1 - v^2/c^2}}$$

$$\nu = \frac{1}{T} = \frac{1}{t}\Big(1 + \frac{v}{c}\Big)^{-1} = \frac{\sqrt{1 - v^2/c^2}}{t_0\,(1 + v/c)} = \nu_0\,\frac{(1 - \frac{v}{c})^{1/2}(1 + \frac{v}{c})^{1/2}}{(1 + \frac{v}{c})}$$

$$\boxed{\nu = \nu_0\sqrt{\frac{1 - v/c}{1 + v/c}}} \qquad \nu < \nu_0, \quad \lambda > \lambda_0 \;\Rightarrow\; \textbf{red shift}$$

16.3 Case 2: source approaching the observer (blue shift)

Now the source has moved *closer* by vt, so the extra travel time is subtracted:

$$T = t - \frac{vt}{c} = t\Big(1 - \frac{v}{c}\Big),$$

$$\nu = \frac{1}{T} = \frac{1}{t}\Big(1 - \frac{v}{c}\Big)^{-1} = \frac{\sqrt{1 - v^2/c^2}}{t_0(1 - v/c)}.$$

$$\boxed{\nu = \nu_0\sqrt{\frac{1 + v/c}{1 - v/c}}} \qquad \nu > \nu_0, \quad \lambda < \lambda_0 \;\Rightarrow\; \textbf{blue shift}$$

Intuition

Two effects multiply: (i) *classical* Doppler, the change in the arrival spacing because the source moves along the line of sight, giving $(1 \pm \beta)^{\mp 1}$; (ii) *time dilation*, the universal factor $\sqrt{1-\beta^2} = \sqrt{(1-\beta)(1+\beta)}$. Combined, $\sqrt{1-\beta^2}/(1 \pm \beta) = \sqrt{(1 \mp \beta)/(1 \pm \beta)}$. Note that the two cases are related by $v \to -v$, and that the product of the receding and approaching factors is exactly 1.

Quick check. $v \ll c$: $\nu \approx \nu_0(1 \mp v/c)$, the ordinary Doppler formula. Receding with $v \to c$: $\nu \to 0$. Approaching with $v \to c$: $\nu \to \infty$.

Same result from four-momentum. A photon moving along $+x$ has $E/c = p_x = h\nu/c$. Boost to a frame S' moving with $+v$ along the photon's direction (an observer running away from the source, equivalent to recession): $E' = \gamma(E - \beta\, pc) = \gamma(1-\beta)E = E\sqrt{\frac{1-\beta}{1+\beta}}$, i.e. the red-shift formula, obtained in one line.

Note on the handwritten notes

In the handwritten page for Case 1, the left side of the boxed result looks like ν_0. It should be ν (the observed frequency), otherwise it would read $\nu_0 = \nu_0\sqrt{\cdots}$.

17 Lorentz Transformations as a Group of Symmetries

17.1 Rotations first: the orthogonal groups

An ordinary rotation $x_i' = R_{ij}x_j$ leaves the Euclidean length unchanged. In 2D, this requires $R^T R = I$ (an *orthogonal* transformation).

Group	Invariant	Meaning
$SO(2)$	$x^2 + y^2 = r^2$	special orthogonal group in 2 dimensions
$SO(3)$	$x^2 + y^2 + z^2 = r^2$	special orthogonal group in 3 dimensions
$SO(n)$	$x_1^2 + x_2^2 + \cdots + x_n^2$	rotations in n dimensions

"Special" means $\det R = +1$ (no reflections).

17.2 Lorentz transformations: rotations with a different metric

A Lorentz transformation acts on any four-vector as

$$A^\mu \to A'^\mu = \Lambda^\mu{}_\nu A^\nu, \qquad x^\mu \to x'^\mu = \Lambda^\mu{}_\nu x^\nu,$$

where $\Lambda^\mu{}_\nu$ is the *Lorentz matrix*. It is chosen so that the invariant quantity is preserved:

$$x'^\mu x'_\mu = x^\mu x_\mu = c^2\mathrm{d}t^2 - \mathrm{d}x^2 - \mathrm{d}y^2 - \mathrm{d}z^2 \quad \text{(a Lorentz scalar)}.$$

(With the opposite signature the same quantity is $-c^2\mathrm{d}t^2 + \mathrm{d}x^2 + \mathrm{d}y^2 + \mathrm{d}z^2$; the group is $SO(3,1)$ either way.)

Signature	$g_{\mu\nu}$	Used for
$(-,+,+,+)$	$\mathrm{diag}(-1,+1,+1,+1)$	$ds^2 = -c^2\mathrm{d}t^2 + \mathrm{d}\vec{x}^2$
$(+,-,-,-)$	$\mathrm{diag}(+1,-1,-1,-1)$	$ds^2 = c^2\mathrm{d}t^2 - \mathrm{d}\vec{x}^2$ (used in these notes)

Intuition

$SO(4)$ would preserve $x_0^2 + x_1^2 + x_2^2 + x_3^2$ (all plus signs): a true 4D rotation. Time enters the Minkowski invariant with the *opposite* sign, so the symmetry group is $SO(3,1)$, which is **not** $SO(4)$. The minus sign is what makes boosts hyperbolic instead of circular, and what makes a light cone exist.

17.3 The defining condition on Λ

Start from invariance of the scalar and use the metric to lower indices:

$$x'^{\mu}x'_{\mu} = g_{\mu\nu}x'^{\mu}x'^{\nu} = g_{\mu\nu}\,\Lambda^{\mu}{}_{\alpha}x^{\alpha}\,\Lambda^{\nu}{}_{\beta}x^{\beta}$$
$$\stackrel{!}{=} x^{\alpha}x_{\alpha} = g_{\alpha\beta}x^{\alpha}x^{\beta}.$$

Since x^{α} is arbitrary,

$$\boxed{g_{\mu\nu}\Lambda^{\mu}{}_{\alpha}\Lambda^{\nu}{}_{\beta} = g_{\alpha\beta}} \iff \Lambda^T g \Lambda = g.$$

The notes also give the equivalent inverse-metric form (apply to $g^{\alpha\beta}$):

$$g^{\alpha\beta}\Lambda^{\mu}{}_{\alpha}\Lambda^{\nu}{}_{\beta} = g^{\mu\nu} \iff \Lambda g \Lambda^T = g$$

(the notes' $\Lambda g \Lambda^T = g$). Both are the Lorentz analogue of $R^T R = I$: replace I by the metric g. For ordinary rotations $g = I$ and this reduces to orthogonality.

17.4 Determinant of Λ

Take the determinant of $\Lambda g \Lambda^T = g$:

$$\det\Lambda\,\det g\,\det\Lambda^T = \det g \;\Rightarrow\; (\det\Lambda)^2 = 1 \;\Rightarrow\; \boxed{\det\Lambda = \pm 1}.$$

- $\det\Lambda = +1$: **proper** Lorentz transformation (boosts, rotations, and products of them).
- $\det\Lambda = -1$: **improper** Lorentz transformation (involves a reflection such as parity).

Extra (not in the class notes): there is a second discrete choice, $\Lambda^0{}_0 \geq 1$ (orthochronous, preserves the direction of time) versus $\Lambda^0{}_0 \leq -1$. The "proper orthochronous" part is the one continuously connected to the identity, which is what ordinary boosts and rotations give.

Same proof in the $x_0 = ict$ convention. Write $X = (x_0, x_1, x_2, x_3)^T$ with $x_0 = ict$, so $ds^2 = \mathrm{d}x_0^2 + \mathrm{d}x_1^2 + \mathrm{d}x_2^2 + \mathrm{d}x_3^2 = \mathrm{d}X^T\mathrm{d}X$ (all plus signs, because $(ict)^2 = -c^2t^2$). Then $X' = \Lambda X$ gives

$$ds'^2 = (\Lambda \mathrm{d}X)^T(\Lambda \mathrm{d}X) = \mathrm{d}X^T\Lambda^T\Lambda\mathrm{d}X \stackrel{!}{=} \mathrm{d}X^T\mathrm{d}X \;\Rightarrow\; \Lambda^T\Lambda = I \;\Rightarrow\; (\det\Lambda)^2 = 1.$$

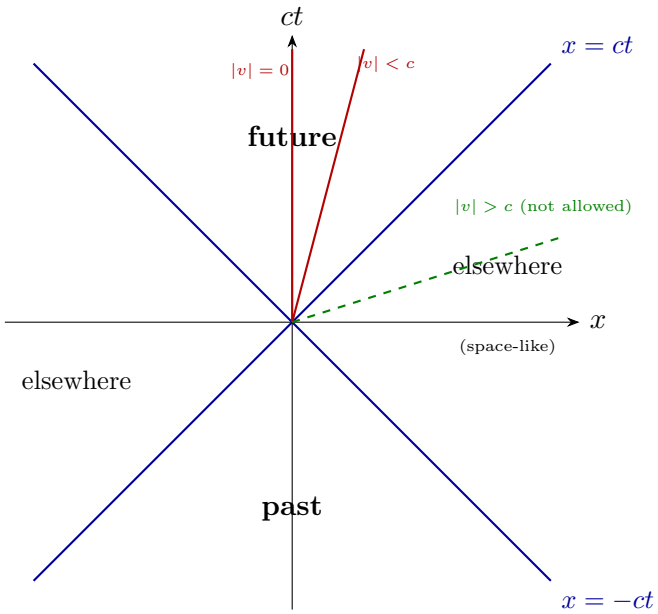
So in this convention a Lorentz transformation is literally an orthogonal matrix, but with complex entries, which is the content of the next sections.

18 Spacetime Interval and the Light Cone

$$ds^2 = c^2\mathrm{d}t^2 - \mathrm{d}\vec{x}^2 \quad (\text{signature } +---) \qquad \left(\text{notes: } ds^2 = -c^2\mathrm{d}t^2 + \mathrm{d}\vec{x}^2\right)$$

Because it is invariant, the *sign* of ds^2 is an absolute classification of the separation between two events.

Condition	Name	ds^2 (in $+---$)	Can they be causally connected?
$c\,\mathrm{d}t = \mathrm{d}x$	light-like (null)	$= 0$	only by a light signal
$c\,\mathrm{d}t > \mathrm{d}x$	time-like	> 0	yes (by a slower-than-light signal)
$c\,\mathrm{d}t < \mathrm{d}x$	space-like	< 0	no



- $|v| = 0$: worldline along the ct axis. $|v| < c$: steeper than the light lines. $|v| = c$: exactly on the lines $ct = \pm x$. $|v| > c$: shallower than the light lines, which would lie in the space-like region (forbidden for matter and signals).
- The two cones about the origin (swept out by rotating the $ct = \pm x$ lines around the t-axis in the 3D picture) are the **future** (upper) and **past** (lower) light cones. Events inside are time-like separated from the origin.
- The region outside the cones (including the $t = 0$ plane that the notes label "present") is space-like separated from the origin: it is the region **"elsewhere"**. Strictly, "present" is not a frame-independent notion; a different observer's "now" is a different tilted plane lying entirely within the elsewhere region.

Intuition

The light cone is the same for all inertial observers because c is the same for all of them. Ordering of time-like separated events is also absolute: they cannot be swapped by any boost. For space-like separated events, one observer sees A before B, another sees B before A. That is the relativity of simultaneity, and it is safe because no cause can reach from A to B.

19 The Boost Matrix as an Imaginary Rotation

With $x_0 = ict$ and $X = (x_0, x_1, x_2, x_3)^T$, a boost along x is

$$\Lambda^{\mu}{}_{\nu} = \begin{pmatrix} \gamma & -i\beta\gamma & 0 & 0 \\ i\beta\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Check against the standard transformation. $x_0' = \gamma x_0 - i\beta\gamma x_1$ means $ict' = \gamma\, ict - i\beta\gamma x \Rightarrow t' = \gamma(t - \beta x/c)$. And $x_1' = i\beta\gamma x_0 + \gamma x_1 = -\beta\gamma ct + \gamma x = \gamma(x - vt)$. Both match Section 5.

Identify an angle. Compare with the ordinary rotation matrix

$$\begin{pmatrix} \cos\theta & -\sin\theta & 0 & 0 \\ \sin\theta & \cos\theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Matching entries gives

$$\cos\theta = \gamma = \frac{1}{\sqrt{1 - v^2/c^2}}, \qquad \sin\theta = i\beta\gamma, \qquad \tan\theta = i\beta = \frac{iv}{c}.$$

Consistency: $\cos^2\theta + \sin^2\theta = \gamma^2(1-\beta^2) = 1$. ✓

Intuition

A boost is a rotation in the (x_0, x_1) plane by an *imaginary* angle. Writing $\theta = i\phi$ turns the circular functions into hyperbolic ones: $\cos\theta = \cosh\phi = \gamma$, $\sin\theta = i\sinh\phi = i\beta\gamma$, $\tanh\phi = \beta$. The number ϕ is the **rapidity**; in the real (ct, x) basis the boost is $\begin{pmatrix} \cosh\phi & -\sinh\phi \\ -\sinh\phi & \cosh\phi \end{pmatrix}$. Rapidities add under successive collinear boosts, which is exactly why velocities obey Einstein's addition rule: $\tanh(\phi_1 + \phi_2) = \dfrac{\beta_1 + \beta_2}{1 + \beta_1\beta_2}$.

19.1 How many Lorentz transformations are there?

- 3 independent **boosts** (one along each of x, y, z) and 3 independent **rotations** (about each axis) constitute the **Lorentz group**: 6 parameters in all.
- Counting check: a general 4×4 matrix has 16 entries; the condition $\Lambda^T g \Lambda = g$ is a symmetric 4×4 equation, i.e. 10 constraints, leaving $16 - 10 = 6$. In general $SO(n)$ has $n(n-1)/2$ parameters; $n = 4$ gives 6.
- $SO(3,1) \neq SO(4)$: invariants $-x_0^2 + x_1^2 + x_2^2 + x_3^2 = r^2$ versus $x_0^2 + x_1^2 + x_2^2 + x_3^2 = r^2$.

20 Exercise: $SO(2)$ from an Exponential, and the Idea of a Generator

Consider the 2D rotation by angle θ:

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$

Claim: $R = e^{i\theta\sigma_2}$ with the Pauli matrix

$$\sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \sigma_2^2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \mathbb{1}.$$

Expand the exponential, using $\sigma_2^2 = \mathbb{1}$ (so $\sigma_2^3 = \sigma_2$, $\sigma_2^4 = \mathbb{1}$,...):

$$\begin{aligned} e^{i\theta\sigma_2} &= \mathbb{1} + i\theta\sigma_2 - \frac{\theta^2}{2!}\mathbb{1} - i\frac{\theta^3}{3!}\sigma_2 + \frac{\theta^4}{4!}\mathbb{1} + \cdots \\ &= i\sigma_2\Big(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \cdots\Big) + \mathbb{1}\Big(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \cdots\Big) \\ &= \mathbb{1}\cos\theta + i\sigma_2\sin\theta. \end{aligned}$$

Since $i\sigma_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, this is $\begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}$.

Note on the handwritten notes

This result is the rotation by $-\theta$, i.e. the transpose of the matrix drawn on the page. To get exactly $\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ use $R = e^{-i\theta\sigma_2}$ (equivalently, rotate by $\theta \to -\theta$). It is only a sign convention, but worth noting if this exercise is asked in an exam. (Interestingly, the $(x', y') = \begin{pmatrix} \cos\phi & \sin\phi \\ -\sin\phi & \cos\phi \end{pmatrix}(x, y)$ form in the tensor pages is precisely the one $e^{i\phi\sigma_2}$ produces.)

σ_2 is the **generator** of the rotation: the whole finite rotation is built from one fixed matrix, and the parameter θ just says how far to go. The general idea "group element = exp(parameter × generator)" carries over to Lorentz transformations:

Transformation	Generator G	Result of $e^{\text{param}\cdot G}$
Rotation by θ (in plane)	$i\sigma_2$, with $(i\sigma_2)^2 = -\mathbb{1}$	$\cos\theta\,\mathbb{1} + (i\sigma_2)\sin\theta$
Boost with rapidity ϕ	$K = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ $(K^2 = +\mathbb{1})$	$\cosh\phi\,\mathbb{1} + K\sinh\phi$

The only difference between $SO(2)$ and a boost is the sign of the square of the generator (-1 versus $+1$), which is the algebraic origin of "trig versus hyperbolic".

21 Worked Example: Drawing the S' Grid on a Spacetime Diagram

Problem. An inertial frame S' moves in the x-direction at $v = \frac{3}{5}c$ relative to S. (i) Draw the lines representing $x' = -3, -2, -1, 0, 1, 2, 3$. (ii) Draw the lines representing $ct' = -3, -2, -1, 0, 1, 2, 3$.

Setup. $\beta = \frac{3}{5}$, $\gamma = \dfrac{1}{\sqrt{1 - 9/25}} = \dfrac{5}{4}$.

(i) Lines of constant x'

$$x' = \gamma(x - \beta ct) = \frac{5}{4}\Big(x - \frac{3}{5}ct\Big) = \frac{5x - 3ct}{4}.$$

Setting $x' = n$: $5x - 3ct = 4n$, i.e.

$$\boxed{ct = \frac{5x - 4n}{3} = \frac{5}{3}x - \frac{4n}{3}}$$

These are parallel lines with slope $dct/dx = 5/3$ (i.e. $dx/d(ct) = 3/5 = \beta$: each one is the worldline of a point fixed in S'). For $n = -3$: $ct = \frac{5}{3}x + 4$, which crosses the ct-axis at $ct = 4$ (i.e. $t = 4/c$) and the x-axis at $x = -\frac{12}{5}$. For $n = 0$: $ct = \frac{5}{3}x$, which is the ct'-axis.

(ii) Lines of constant ct'

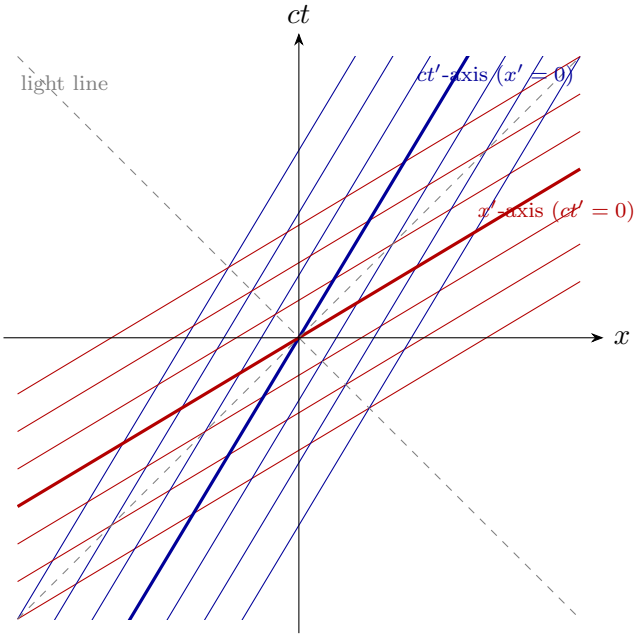
$$ct' = \gamma(ct - \beta x) = \frac{5}{4}\Big(ct - \frac{3}{5}x\Big) = \frac{5ct - 3x}{4}.$$

Setting $ct' = n$: $5ct - 3x = 4n$, so

$$\boxed{ct = \frac{3x + 4n}{5} = \frac{3}{5}x + \frac{4n}{5}}$$

Slope $3/5 = \beta$ (tilted *up* by the same amount that the ct' axis is tilted *in*). For $n = -3$: $ct = \frac{3}{5}x - \frac{12}{5}$, crossing the x-axis at $x = 4$ and the ct-axis at $ct = -\frac{12}{5}$ (i.e. $t = -\frac{12}{5c}$). For $n = 0$: $ct = \frac{3}{5}x$, which is the x'-axis.

n	-3	-2	-1	0	1	2	3
line $x' = n$: $ct = \frac{5x-4n}{3}$, ct-intercept	4	$\frac{8}{3}$	$\frac{4}{3}$	0	$-\frac{4}{3}$	$-\frac{8}{3}$	-4
line $ct' = n$: $ct = \frac{3x+4n}{5}$, ct-intercept	$-\frac{12}{5}$	$-\frac{8}{5}$	$-\frac{4}{5}$	0	$\frac{4}{5}$	$\frac{8}{5}$	$\frac{12}{5}$



Blue: lines of constant x' (worldlines of points fixed in S'; slope $5/3$ in ct vs. x, i.e. speed $\frac{3}{5}c < c$, so steeper than the light lines). Red: lines of constant ct' (simultaneity lines of S').

Note on the handwritten notes

In the handwritten solution for (i) the ct-intercept of the line $x' = -3$ is labelled $12/c$. From $5x - 3ct = -12$ at $x = 0$ one gets $ct = 4$, i.e. $t = 4/c$. The x-intercept $-12/5$ in the page is correct. In (ii) both intercepts ($x = 4$ and $t = -12/(5c)$) match.

Intuition

The two axes of S' tilt toward the light line symmetrically: ct'-axis leans toward $x = ct$ with slope $1/\beta$, the x'-axis leans up with slope β. They squeeze together as $v \to c$ and merge with the light line. The "tick marks" $n = \pm1, \pm2, \pm3$ are not equally spaced *as seen on the page* by Euclidean measure, because the invariant interval, not the Euclidean distance, defines unit length: the point $(x', ct') = (1, 0)$ sits at $(x, ct) = (\gamma, \gamma\beta) = (\frac{5}{4}, \frac{3}{4})$ in S, a Euclidean distance > 1 from the origin, yet $ds^2 = \gamma^2 - \gamma^2\beta^2 = 1$.